Does Width or Depth Matter More in Multilayer Perceptrons? An Experimental Study of Performance and Generalisation

https://github.com/Vinaykumar3678/mlp-depth-vs-width-tutorial

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ABSTRACT

Multilayer Perceptrons (MLPs) are widely used neural networks for classification tasks. A key design decision when constructing an MLP is choosing the network architecture, particularly the number of hidden layers (depth) and the number of neurons per layer (width). These factors directly influence model capacity, training efficiency, and generalisation performance.

This tutorial investigates how depth and width affect MLP performance using the Fashion-MNIST dataset. Controlled experiments were conducted in which network architectures were varied while keeping all other parameters constant. The models were evaluated using accuracy, loss curves, generalisation gap, and confusion matrices.

The results demonstrate that increasing width improves performance more effectively than increasing depth. Wider networks achieve higher accuracy and better generalisation, while deeper networks show diminishing returns under the given conditions.

1. INTRODUCTION

Artificial neural networks are widely used for solving complex classification problems. Among these, Multilayer Perceptrons (MLPs) are one of the simplest yet effective architectures.

The structure of an MLP is defined by:

- **Depth**: number of hidden layers
- **Width**: number of neurons per layer

These parameters significantly influence how well the model learns patterns in data (Goodfellow et al., 2016).

While deeper networks can theoretically learn more complex hierarchical representations, they are often harder to train due to optimisation challenges. Wider networks, on the other hand, increase representational capacity without significantly increasing training complexity.

This tutorial aims to answer:

2. BACKGROUND

Multilayer Perceptrons consist of layers of neurons where each neuron performs a weighted sum of inputs followed by a non-linear activation function. Training is performed using backpropagation, which adjusts weights by minimising a loss function (Rumelhart et al., 1986).

Key theoretical insights include:

- Deep networks learn hierarchical feature representations
- Wide networks increase the expressive capacity of each layer
- According to the Universal Approximation Theorem, sufficiently wide networks can approximate any continuous function (Cybenko, 1989)

However, deeper networks may suffer from:

- vanishing gradients
- increased training time
- optimisation difficulty
- overfitting

3. DATASET AND EXPERIMENTAL SETUP

Dataset

The **Fashion-MNIST dataset** consists of:

- 60,000 training images
- 10,000 test images
- 10 classes of clothing items
- 28×28 grayscale images

It provides a more challenging alternative to the original MNIST dataset (Xiao et al., 2017).

Experimental Design

To ensure a fair comparison, all models were trained under identical conditions:

- Optimiser: Adam
- Epochs: 20
- Batch size: 128

Two sets of experiments were conducted:

Depth Analysis

- Hidden layers: 1, 3, 5
- Width fixed at 128 neurons

Width Analysis

- Neurons per layer: 32, 128, 512
- Depth fixed at 2 layers

4. RESULTS AND VISUALISATION

4.1 Validation Accuracy for Depth-Focused Models

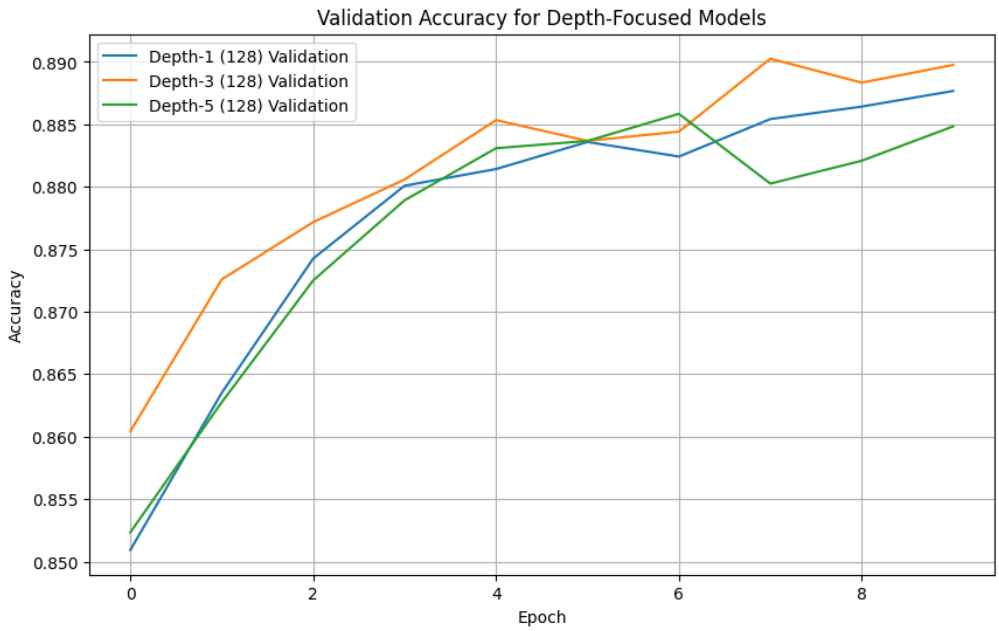
Figure 1: Validation Accuracy for Depth-Focused Models

Figure 1 presents the validation accuracy of the depth-focused models across training epochs. In this experiment, the number of neurons per hidden layer was fixed at 128, while the number of hidden layers was varied between one, three, and five. This comparison was designed to evaluate whether increasing depth improves classification performance.

The graph shows that all three models improve steadily during the early epochs, indicating that each architecture is able to learn meaningful patterns from the Fashion-MNIST dataset. However, the

differences between the models become clearer as training progresses. The three-layer network achieves the highest validation accuracy overall, reaching approximately 0.89, while the one-layer model follows closely behind. The five-layer model performs competitively but shows slightly greater fluctuation in later epochs.

These results suggest that moderate depth provides a useful increase in learning capacity, but adding further layers does not guarantee better performance. Instead, very deep models may introduce training instability without delivering a meaningful improvement in validation accuracy.



4.2 Validation Accuracy for Width-Focused Models

Figure 2: Validation Accuracy for Width-Focused Models

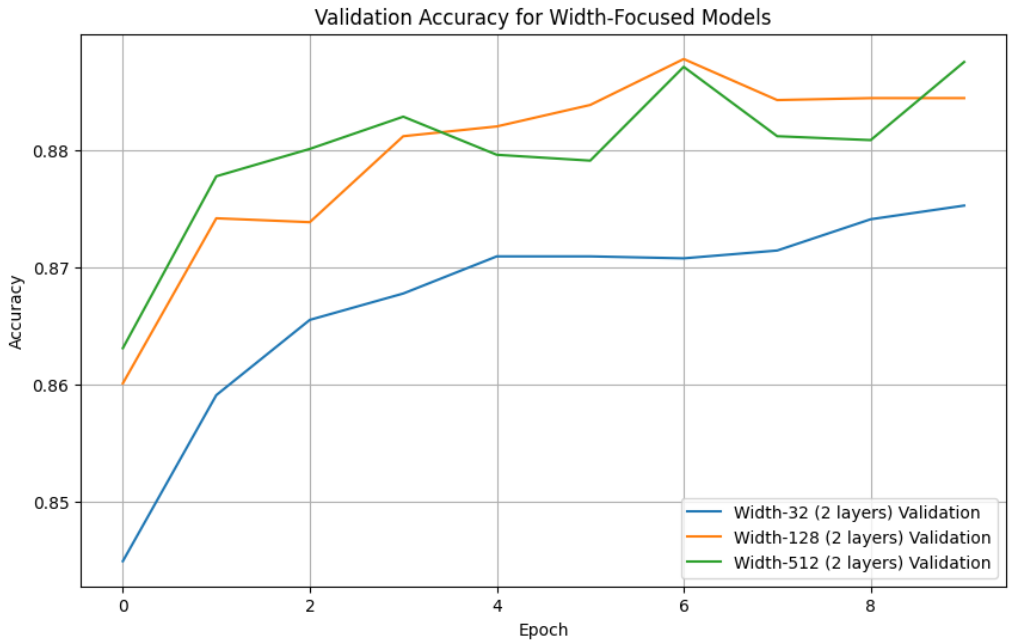


Figure 2 shows the validation accuracy of the width-focused models over the course of training. In this experiment, the number of hidden layers was fixed at two, while the number of neurons per layer was varied between 32, 128, and 512. This allowed the effect of width on model performance to be examined independently of depth.

The graph demonstrates that wider models achieve better validation accuracy and tend to converge more effectively. The model with 512 neurons performs best overall, indicating that increasing the number of neurons improves the model's capacity to capture useful feature representations. The 128-neuron model also performs well, while the 32-neuron model achieves the lowest validation accuracy throughout most of training.

This figure shows that width has a clear and positive impact on learning performance. Compared with the depth-focused results, the effect of increasing width is more noticeable, suggesting that additional neurons are more beneficial than additional layers for this task.

4.3 Final Test Accuracy of All Models

Figure 3: Final Test Accuracy of All Models

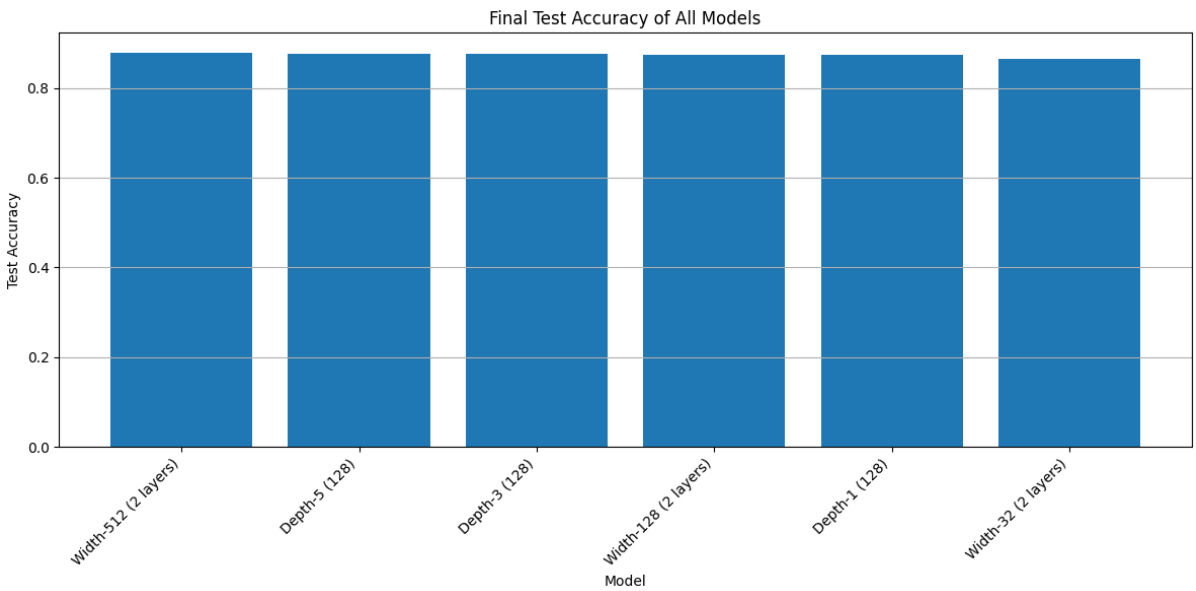


Figure 3 compares the final test accuracy of all six models included in the study. This figure provides the clearest summary of overall model performance because it evaluates each architecture on unseen test data.

The results show that the width-focused models outperform most of the depth-focused models. In particular, the network with 512 neurons achieves the highest test accuracy, making it the best-performing model in the experiment. The three-layer depth-focused model performs well, but it does not surpass the strongest width-focused configuration. The lowest performance is observed in the narrowest model, which indicates that insufficient width limits the model's ability to learn complex patterns.

This figure provides strong evidence that width has a greater effect on final classification accuracy than depth under the chosen experimental conditions. It also supports the conclusion that a wider, moderately deep architecture is the most effective design for this dataset.

4.4 Generalisation Gap of All Models

Figure 4: Generalisation Gap of All Models

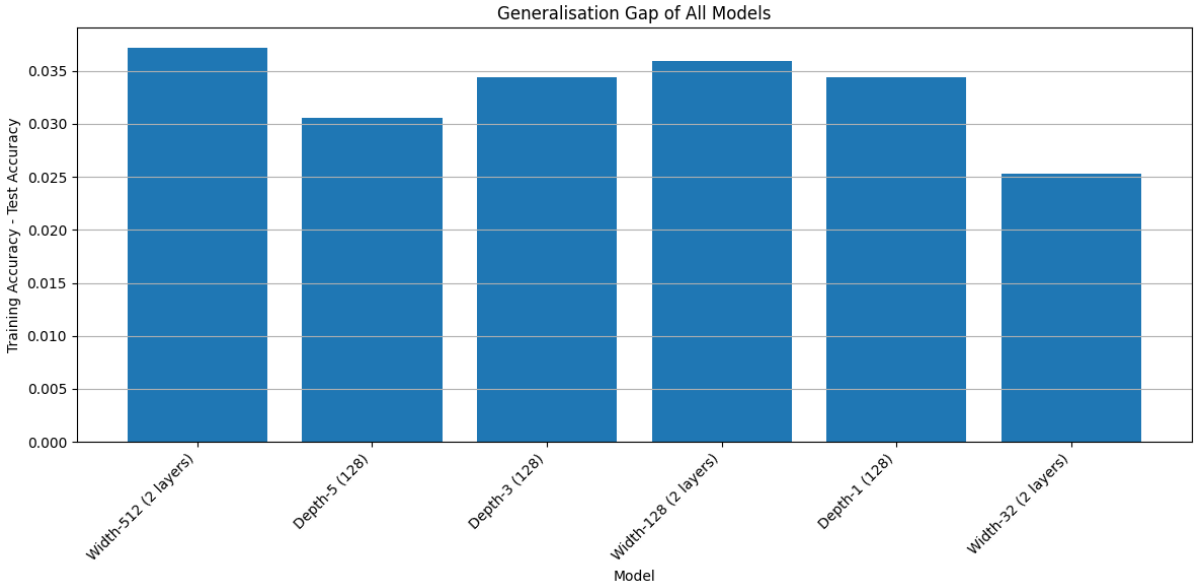


Figure 4 illustrates the generalisation gap for each model. The generalisation gap is calculated as the difference between training accuracy and validation accuracy, and it provides an indication of how well a model generalises to unseen data. A smaller gap generally suggests better generalisation, while a larger gap may indicate overfitting.

The graph shows that some models achieve high training performance but do not maintain the same advantage on validation data. The wider models, particularly the stronger-performing configurations, display relatively small generalisation gaps, suggesting that they learn useful patterns without overfitting excessively. In contrast, some deeper models show slightly larger gaps, indicating that additional depth may increase the risk of fitting the training data more closely without producing a corresponding benefit on unseen examples.

This figure is important because it moves beyond simple accuracy comparison. It shows not only which models perform well, but also which models do so in a way that is more likely to remain reliable on new data.

4.5 Training Time of All Models

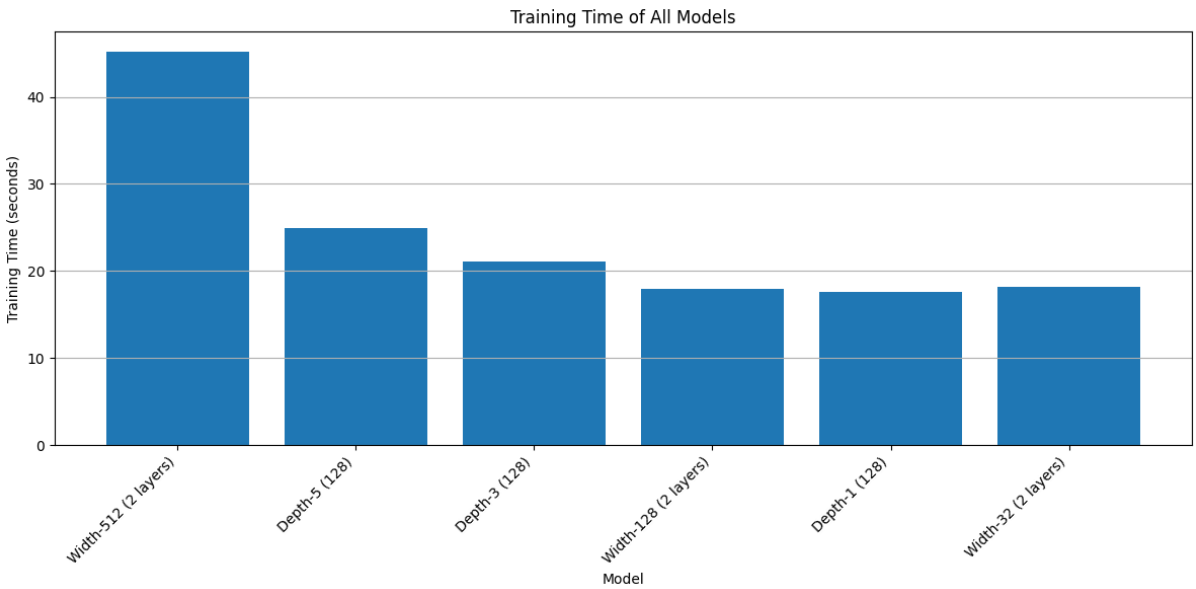
Figure 5: Training Time of All Models

Figure 5 compares the total training time required for each model. This is an important practical consideration because model selection should not depend only on accuracy, but also on computational efficiency.

The figure shows that training time generally increases as model complexity increases. Deeper and wider networks require more computation because they contain more parameters and involve more operations during both forward and backward propagation. The narrowest model trains most quickly, while the largest models require the longest training time. However, the most accurate model also has a higher computational cost.

This result highlights an important trade-off between performance and efficiency. Although wider models provide better predictive accuracy, they also demand more processing time. Therefore, the

choice of architecture should depend on whether the priority is maximum performance or a balance between performance and computational cost.



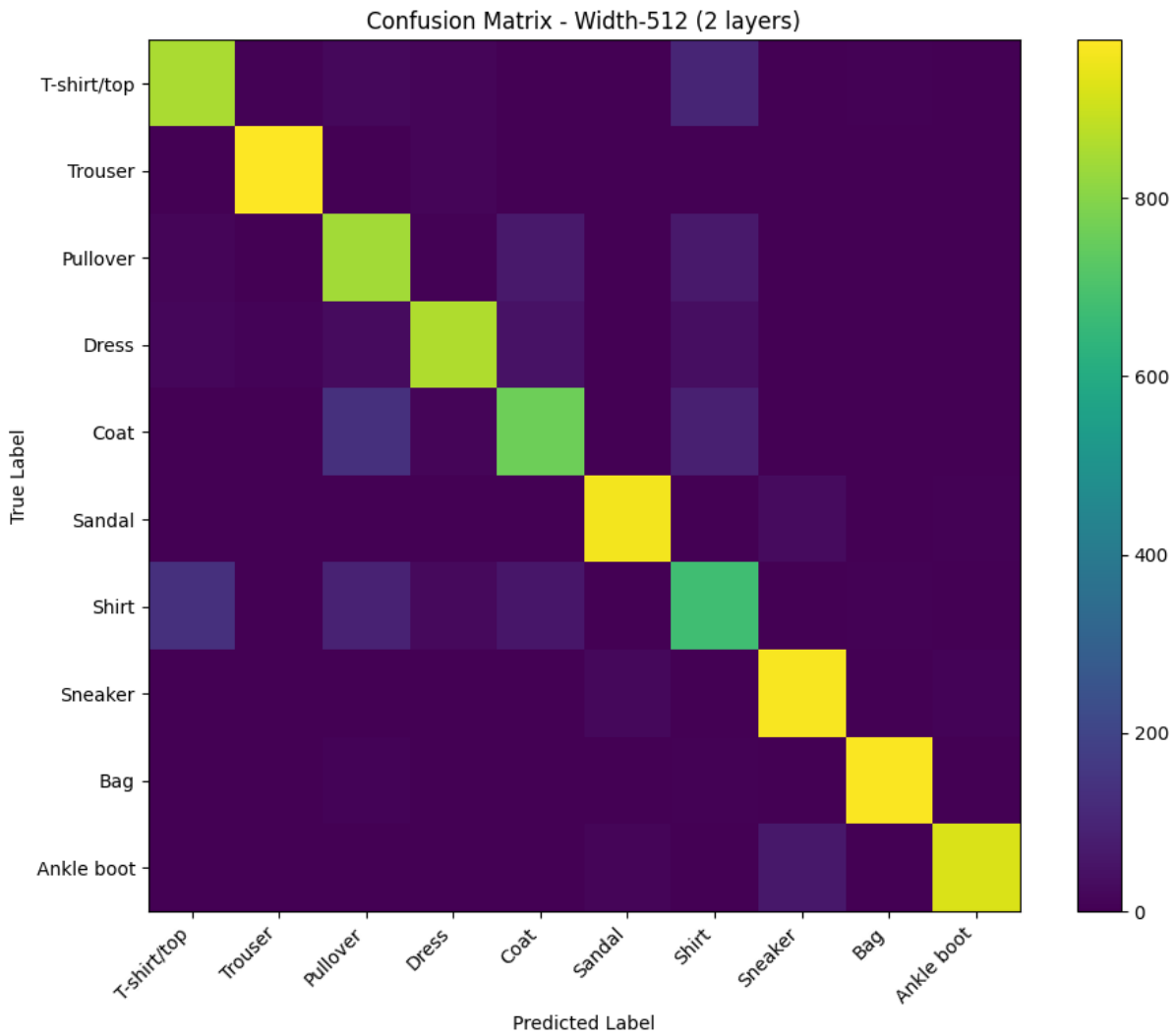
4.6 Confusion Matrix of the Best-Performing Model

Figure 6: Confusion Matrix of the Best-Performing Model

Figure 6 presents the confusion matrix for the best-performing model identified in the experiment. The confusion matrix provides a class-by-class breakdown of predictions and helps to reveal where the model performs well and where errors are more likely to occur.

The matrix shows strong classification performance across several classes, with high values concentrated along the diagonal, which indicates correct predictions. However, some misclassifications are visible between visually similar categories. For example, clothing types with similar shapes or textures are more likely to be confused by the model. This is expected in image classification tasks where some classes share overlapping visual characteristics.

This figure adds a deeper level of evaluation to the report. While earlier figures demonstrate overall performance trends, the confusion matrix explains the nature of the remaining errors and provides a more complete understanding of model behaviour.



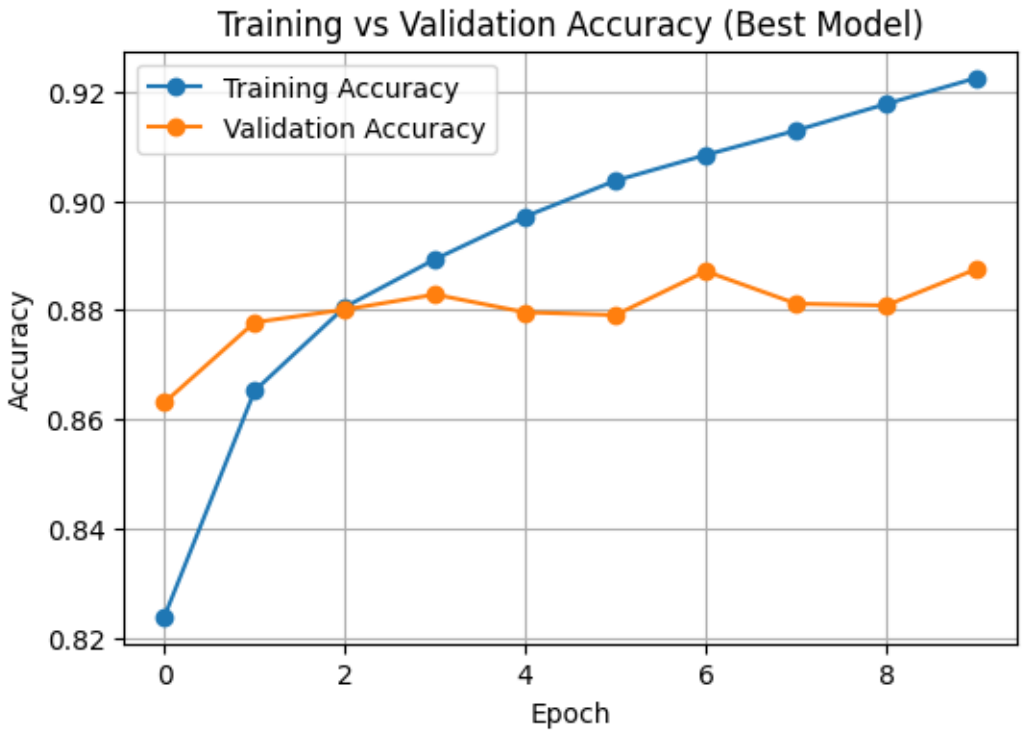
4.7 Training vs Validation Accuracy of the Best Model

Figure 7: Training and validation accuracy for the best-performing model.

This figure shows the training and validation accuracy curves for the best-performing model across training epochs. The training accuracy increases steadily, indicating that the model effectively learns patterns from the training data.

The validation accuracy follows a similar trend and remains close to the training curve throughout the training process. The small gap between the two curves suggests that the model generalises well and does not suffer from significant overfitting.

Overall, this figure supports the earlier findings from the generalisation gap analysis and confirms that the selected model achieves a good balance between learning capacity and generalisation performance.



5. DISCUSSION

The results indicate that network width has a greater impact on performance than depth under the given experimental conditions.

Wider networks improve the model's ability to capture complex patterns by increasing representational capacity within each layer. This leads to faster convergence and higher accuracy. In contrast, deeper networks provide limited improvement and introduce additional training challenges.

These findings suggest that for moderately complex datasets such as Fashion-MNIST, increasing width is more effective than increasing depth.

Limitations

- Limited number of training epochs
- No regularisation techniques applied
- No hyperparameter tuning

Ethical Considerations

- Dataset bias may limit generalisability
- Larger models require more computational resources
- Misclassification could impact real-world applications

6. CONCLUSION

This tutorial investigated the effect of architectural design on MLP performance.

The findings show that:

- Increasing width significantly improves performance
- Increasing depth provides limited benefit
- Effective model design requires balancing performance and computational cost

These results highlight the importance of selecting appropriate neural network architectures in practical applications.

7. ACCESSIBILITY CONSIDERATIONS

- Clear structure and logical organisation
- Readable formatting and consistent headings
- Visualisations designed for clarity

Future improvements could include:

- colour-blind friendly visualisations
- alternative text descriptions for figures
- improved compatibility with screen readers

8. REFERENCES

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